One-head attention pooling adds little beyond a deeper MLP in a Qwen2.5-VL region probe

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https://github.com/nkn002/visual-readout

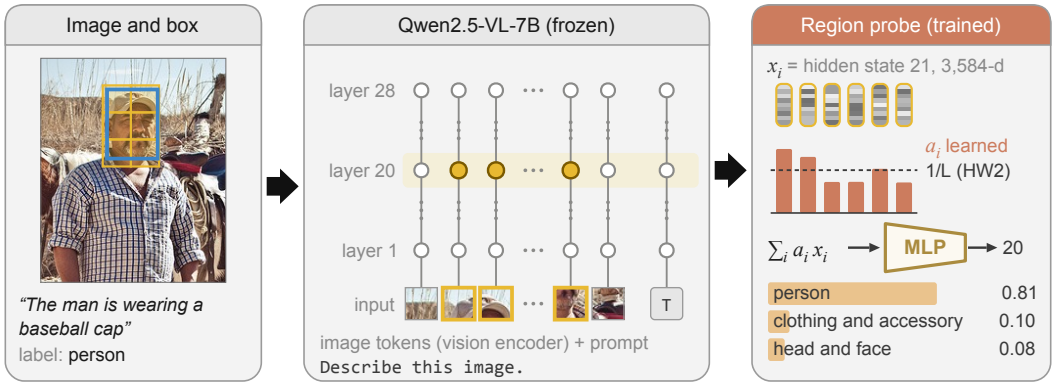


Figure 1: The probe reads the frozen VLM's hidden state 21 at the L tokens under the box, not pixels. HW2 weights them by $1/L$; attention learns a_i (values from the 1-head, 1-hidden probe, seed 0).

1 Hypothesis, committed before training

Problem. The probe names what a Visual Genome box shows (20 categories) from the frozen Qwen2.5-VL-7B's hidden state 21 (3,584-d) at the L image tokens under the box (Fig. 1). The HW2 probe feeds their mean to an MLP ($3584 \to 512 \to \text{ReLU} \to 20$), and its weakest category is person (50% recall), whose boxes often show mostly a shirt, a face, or an arm.

Design. We replace the mean with attention pooling, $\sum_i a_i x_i$ with $a_i = \text{softmax}_i\left(q^\top W_k \hat{x}_i / \sqrt{64}\right)$ over the standardized tokens $\hat{x}_i$, and add a second hidden layer ($3584 \to 512 \to \text{ReLU} \to 512 \to \text{ReLU} \to 20$). The query q starts at zero, so training starts from the exact mean pool; an 8-head variant gives each head its own query and 448-d slice.

Predictions. Learned weights can favor object tokens over background tokens. Before any arm trained (commit `c03e240` at 15:02; first result at 15:37), we predicted: (P1) no change on one-token boxes; (P2) gains for boxes of 10 or more tokens, growing with size; (P3) the largest recall gains for person, limb, and clothing; (P4) a second hidden layer on the mean pool adds overfitting, not accuracy; and (P5) 8 heads match or beat 1 head. We also named failure modes F1 to F4 (Tab. 2).

2 Setup

We cross pooling (mean, 1-head attention, 8-head attention) with depth (1 or 2 hidden layers) on the HW2 split: 50,000 regions, 2,500 per category, split by image into 39,968 train, 5,075 validation, and 4,957 test regions. Every arm searches the same Adam grid (learning rate {3e-4, 1e-3} $\times$ weight decay {1e-4, 1e-3}), keeps its best setting on validation accuracy, and reruns it with 3 seeds. Paired-bootstrap 95% CIs over test regions leave out seed variance, and the 843 one-token test boxes act as a placebo, since attention there returns the token itself.

Table 1: The proposed design (gray) beats HW2 by +0.71 points, but depth alone gives +0.48. Test accuracy (%) is the mean ± std over 3 seeds; Δ is the gain in points with its 95% CI.

Pooling	Hidden	Test acc.	Δ vs. HW2	Δ vs. mean, same depth	Epoch (s)
Mean (HW2)	1	73.62 ± 0.38	–	–	0.09
Mean	2	74.10 ± 0.21	+0.48 [+0.01, +0.95]	–	0.21
Attention, 1 head	1	74.10 ± 0.15	+0.48 [−0.09, +1.06]	+0.48 [−0.09, +1.06]	0.77
Attention, 1 head	2	74.33 ± 0.08	+0.71 [+0.05, +1.32]	+0.24 [−0.40, +0.80]	0.69
Attention, 8 heads	1	74.48 ± 0.17	+0.86 [+0.26, +1.43]	+0.86 [+0.26, +1.43]	1.09
Attention, 8 heads	2	**75.07** ± 0.25	+1.45 [+0.82, +2.06]	+0.97 [+0.40, +1.52]	1.23

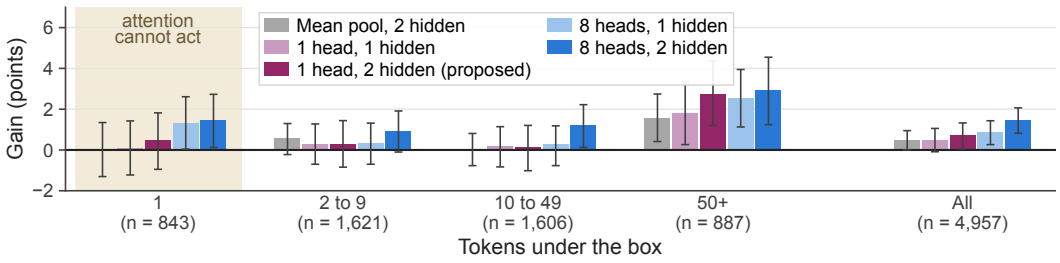


Figure 2: Part of the 8-head gain appears on one-token boxes, where attention cannot act (shaded). Bars: gain over HW2 per token-count bin (3-seed mean), with 95% CIs.

3 Results

The second hidden layer, not one-head attention, carries most of the proposed design's gain. The proposed design beats HW2 by +0.71 points, but the same MLP on the mean pool already gains +0.48, so one-head attention adds only +0.24 at the same depth, with a CI that includes zero (Tab. 1). In contrast, eight heads beat the mean pool at both depths (+0.86 and +0.97), and the best arm reaches 75.07%.

One head gains where predicted, but eight heads also gain where attention cannot act. At HW2's depth, one head gains +0.08 points on one-token boxes and +1.80 on boxes of 50+ tokens (Fig. 2), as P1 and P2 predicted, but depth alone also gains +1.54 there. Interestingly, eight heads at HW2's depth gain +1.30 [+0.04, +2.61] on one-token boxes, where their input equals HW2's, more than their +0.77 on larger boxes. This borderline result (one of about 30 bin contrasts) suggests that part of the eight-head gain comes from the MLP or seed variance, not from re-weighting tokens.

Attention does not fix person recall. Person recall changes by −2.3 points [−5.9, +1.2] for the proposed design and by −0.3 for the best arm. Attention also costs 3 to 12× more time per epoch, and its weights stay diffuse (Tab. 2). Re-scoring the attention probes with uniform weights (not run) would separate the pooling from the MLP.

Table 2: The data support P5, partly support P2 and P4, contradict P1 for eight heads, and do not support P3 for person. Brackets: 95% CIs; F4 (weight decay, outliers) was not tested.

Prediction or failure mode	Observation	Verdict
P1. 1-token boxes unchanged	1 head +0.08; 8 heads +1.30 [+0.04, +2.61]	Only for 1 head
P2. Gain grows with box size	50+ tokens: attention +1.8 to +2.9; depth +1.5	Partly; depth too
P3. Person, limb, clothing	Person −2.3; clothing +3.2 and +6.8	Not for person
P4. Depth: overfitting only	Train 86.4% vs. 82.4%; test +0.48 [+0.01, +0.95]	Partly refuted
P5. 8 heads ≥ 1 head	+0.38 (1 hidden); +0.73 [+0.24, +1.23] (2 hidden)	Supported
F1. Weights stay uniform	Entropy 0.90 to 0.95 (1 is uniform)	No; diffuse
F2. Position lock	First token on top: 15 to 18% vs. 13 to 14%	Weak at most
F3. Overfitting	Train-val. gap 9.4 (HW2), 13.3 (8 heads, 1 hidden)	In one arm